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CPS I — Exercise Sheet 11

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Exercise 1: Safety-Liveness Decomposition

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1. $P_{1safe} = P_1$ ✓
 $P_{1live} = \Sigma^\omega$ ✓
 $P_{1safe} \cap P_{1live} = P_1$
2. $P_{2safe} = \Sigma^\omega$ ✓
 $P_{2live} = P_2$ ✓
 $P_{2safe} \cap P_{2live} = P_2$
3. $P_{3safe} = \{\sigma \in \Sigma^\omega \mid |\{i \in \mathbb{N} \mid a \in A_i\}| \leq 3\}$ (✓)
 $P_{3live} = \underline{P_3}$ - 0.5
 $P_{3safe} \cap P_{3live} = P_3$
4. $P_{4safe} = \{\sigma \in \Sigma^\omega \mid a \in A_0\}$ ✓
 $P_{4live} = \{\sigma \in \Sigma^\omega \mid \forall i \in \mathbb{N} \exists j > i. a \in A_j\}$ ✓
 $P_{4safe} \cap P_{4live} = P_4$
5. $P_{5safe} = \Sigma^\omega$ ✓
 $P_{5live} = \Sigma^\omega$ ✓
 $P_{5safe} \cap P_{5live} = P_5$

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Exercise 2: Satisfaction under Fairness Assumptions

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1. • fair execution: $s_0 \xrightarrow{\alpha} s_4 \xrightarrow{\beta} (s_5 \xrightarrow{\gamma} \cancel{s_5})^\omega$ ✓
 • unfair execution: $(s_0 \xrightarrow{\eta} s_1 \xrightarrow{\eta} s_3 \xrightarrow{\eta} \cancel{s_0})^\omega$ ✓
2. • fair execution: Not exist ✓
 • unfair execution: $s_0 \xrightarrow{\alpha} s_4 \xrightarrow{\beta} (s_5 \xrightarrow{\gamma} \cancel{s_5})^\omega$ ✓
3. • fair execution: $(s_0 \xrightarrow{\alpha} s_4 \xrightarrow{\delta} \cancel{s_0})^\omega$ ✓
 • unfair execution: $(s_0 \xrightarrow{\eta} s_1 \xrightarrow{\eta} s_3 \xrightarrow{\eta} \cancel{s_0})^\omega$ ✓
4. • fair execution: $(s_0 \xrightarrow{\eta} s_1 \xrightarrow{\eta} s_3 \xrightarrow{\eta} \cancel{s_0})^\omega$ (not enabled) ✓

Notation: make sure that states and transitions alternate:

example: $(s_5 \xrightarrow{\gamma} s_5)^\omega$ is the same as

$s_5 \xrightarrow{\gamma} (s_5 \xrightarrow{\gamma} s_5) \xrightarrow{\gamma} (s_5 \xrightarrow{\gamma} s_5) \xrightarrow{\gamma} (s_5 \xrightarrow{\gamma} s_5) \xrightarrow{\gamma} \dots$

does not make sense

- 0.5

- unfair execution: $(s_0 \xrightarrow{\alpha} s_4 \xrightarrow{\delta} \cancel{s_0})^\omega$ ✓
5. • fair execution: $s_0 \xrightarrow{\eta} (s_1 \xrightarrow{\delta} s_2 \xrightarrow{\delta} \cancel{s_1})^\omega$ (not enabled) ✓
- unfair execution: $(s_0 \xrightarrow{\alpha} s_4 \xrightarrow{\delta} \cancel{s_0})^\omega$ ✓
6. • fair execution: $s_0 \xrightarrow{\alpha} s_4 \xrightarrow{\beta} (s_5 \xrightarrow{\gamma} \cancel{s_5})^\omega$ (not enabled) ✓
- unfair execution: $(s_0 \xrightarrow{\alpha} s_4 \xrightarrow{\delta} \cancel{s_0})^\omega$ ✓
7. • fair execution: $(s_0 \xrightarrow{\eta} s_1 \xrightarrow{\eta} s_3 \xrightarrow{\eta} \cancel{s_0})^\omega$ ✓
- unfair execution: $s_0 \xrightarrow{\eta} (s_1 \xrightarrow{\delta} s_2 \xrightarrow{\delta} \cancel{s_1})^\omega$ ✓
8. • fair execution: $s_0 \xrightarrow{\alpha} s_4 \xrightarrow{\beta} (s_5 \xrightarrow{\gamma} \cancel{s_5})^\omega$ (not enabled) ✓
- unfair execution: $s_0 \xrightarrow{\eta} (s_1 \xrightarrow{\delta} s_2 \xrightarrow{\delta} \cancel{s_1})^\omega$ ✓

0/4 (1) and (2) satisfy property "eventually a"

explanation missing, rest missing